Lecture Note: Introduction to the 8085 Microprocessor

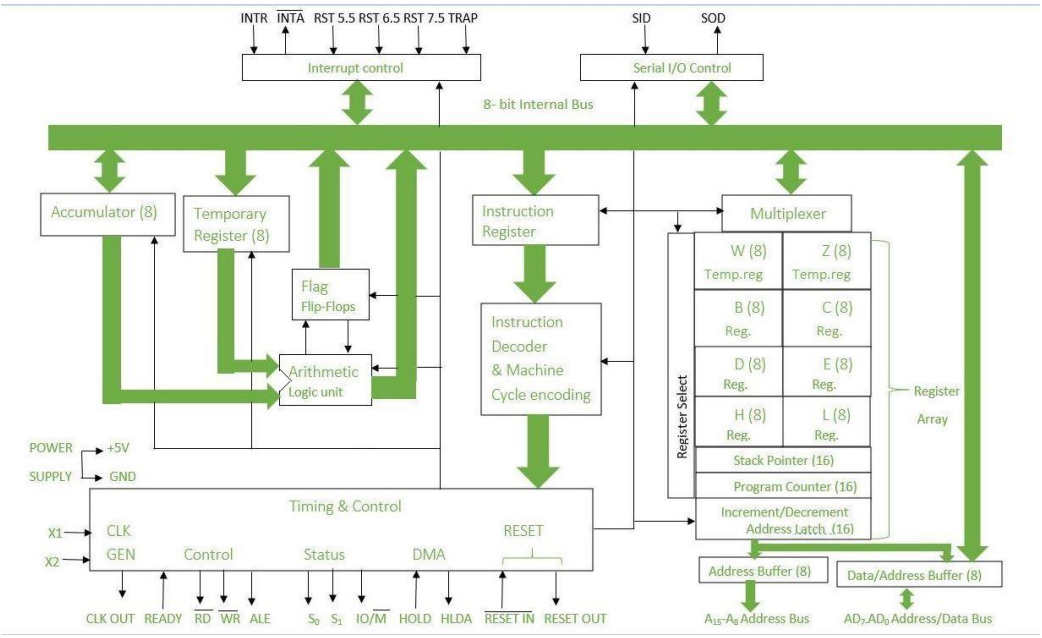
1. Overview of Microprocessors

A microprocessor is a programmable device that fetches, decodes, and executes instructions. It serves as the central processing unit (CPU) of a computer system. The Intel 8085 is an 8-bit microprocessor introduced in 1976. It was widely used in early microcomputers and embedded systems due to its simplicity and cost-effectiveness.

2. Features of the 8085 Microprocessor

- **8-bit Data Bus** : The 8085 can process 8 bits of data at a time.
- **16-bit Address Bus** : It can address up to 216=65,536 memory locations (64 KB).
- **Clock Speed** : Operates at a clock frequency of 3 MHz (up to 5 MHz in some versions).
- **Power Supply** : Requires a single +5V DC power supply.
- **Instruction Set** : Supports 74 basic instructions and 246 opcodes.
- **Registers** : Contains six general-purpose registers (B, C, D, E, H, L), a program counter (PC), a stack pointer (SP), and an accumulator (A).
- **Interrupts** : Provides five hardware interrupts (TRAP, RST 7.5, RST 6.5, RST 5.5, and INTR).
- **Serial I/O** : Includes SID (Serial Input Data) and SOD (Serial Output Data) pins for serial communication.

3. Architecture of the 8085 Microprocessor



The architecture of the 8085 can be divided into three main units:

3.1. Arithmetic and Logic Unit (ALU)

- Performs arithmetic operations (addition, subtraction) and logical operations (AND, OR, XOR, NOT).
- Works with the Accumulator (A register) to store intermediate results.

3.2. Registers

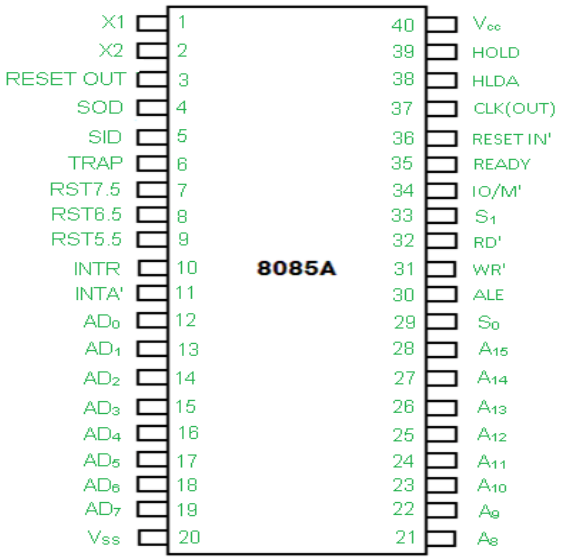
The 8085 has several registers:

- **Accumulator (A)** : An 8-bit register used for arithmetic and logic operations.
- **General-Purpose Registers (B, C, D, E, H, L)** : Used for storing data temporarily.
- **Program Counter (PC)** : A 16-bit register that holds the address of the next instruction to be executed.
- **Stack Pointer (SP)** : A 16-bit register that points to the top of the stack in memory.
- **Temporary Registers (W, Z)** : Internal registers not accessible to the programmer.

3.3. Control Unit

- Generates control signals to coordinate the execution of instructions.
- Manages the timing and sequencing of operations.

4. Pin Diagram of the 8085



The 8085 has 40 pins arranged in a dual in-line package (DIP). Key pins include:

- **Address/Data Bus (AD0-AD7)** : Multiplexed address and data lines.
- **Address Bus (A8-A15)** : Higher-order address lines.
- **Control Signals** :
    - *RD*: Read signal (active low).
    - *WR*: Write signal (active low).
    - *ALE*: Address Latch Enable.
    - *IO*/*M*: Indicates whether the operation is I/O or memory-related.
- **Interrupts** : TRAP, RST 7.5, RST 6.5, RST 5.5, INTR.
- **Serial I/O** : SID (Serial Input Data), SOD (Serial Output Data).

5. Workflow of the 8085 Microprocessor

i. Power Supply and Clock Initialization

- The process begins with the **Power Supply** providing +5V and GND to the microprocessor.
- The **X1, X2** pins receive an external clock signal (CLK GEN), which is used to generate the internal clock signal (CLK OUT) through the **Timing & Control** unit. This unit synchronizes all operations within the microprocessor.
- The **RESET IN** signal can reset the microprocessor, setting the **Program Counter (PC)** to 0000H, while **RESET OUT** indicates that a reset has occurred.

ii. Instruction Fetch

- The **Program Counter (PC)** (16-bit) holds the address of the next instruction to be fetched. It sends this address to the **Address Latch** (16-bit), which splits it into two 8-bit parts for the 8-bit **Address Bus**.
- The **Address Buffer** outputs the address to the external **Address Bus (A0-A15)**, while the **ALE (Address Latch Enable)** signal indicates that the address is valid on the bus.
- The microprocessor asserts the **RD (Read)** signal to fetch the instruction from memory via the **Data/Address Bus (AD0-AD7)**, which is multiplexed to handle both address and data.
- The fetched instruction is placed in the **Instruction Register** (8-bit) via the **8-bit Internal Bus**.

iii. Instruction Decode

- The instruction in the **Instruction Register** is sent to the **Instruction Decoder & Machine Cycle Encoding** unit.

- This unit decodes the instruction and determines the operation to be performed (e.g., arithmetic, logical, data transfer, etc.).
- It also generates the necessary control signals for the **Timing & Control** unit to manage the machine cycles (e.g., opcode fetch, memory read/write, I/O operations).

iv. Operand Fetch and Execution

- If the instruction requires operands (e.g., data or addresses), the 8085 uses the **Address Latch** and **Program Counter** to fetch additional bytes from memory.
- The operands are loaded into the appropriate registers:
    - **General Purpose Registers (B, C, D, E, H, L)**: These 8-bit registers store data temporarily. H and L can be combined to form a 16-bit address for memory operations.
    - **Temporary Register**: Used for intermediate storage during operations.
    - **W and Z Registers**: Temporary 8-bit registers used during specific operations like CALL or memory addressing.
- The **Arithmetic Logic Unit (ALU)** performs the required operation (e.g., addition, subtraction, AND, OR) using data from the **Accumulator** (8-bit) and another register or memory.
- The **Flag Flip-Flops** (e.g., Zero, Sign, Parity, Carry, Auxiliary Carry) are updated based on the result of the ALU operation.

v. Data Transfer and Storage

- The result of the operation is stored in the **Accumulator** or another register, depending on the instruction.
- If the instruction involves memory or I/O operations, the **Data/Address Buffer** is used to send or receive data via the **Data/Address Bus (AD0-AD7)**.
- The **WR (Write)** signal is asserted for writing data to memory or an I/O device.
- The **Stack Pointer (SP)** (16-bit) is used for stack operations (e.g., PUSH, POP, CALL, RET), managing the stack in memory.

vi. Interrupt Handling

- The 8085 supports multiple interrupt signals: **INTR, INTA, RST 5.5, RST 6.5, RST 7.5, TRAP**. These are managed by the **Interrupt Control** unit.
- When an interrupt occurs, the microprocessor completes the current instruction, saves the current **Program Counter** value on the stack, and jumps to the interrupt service routine (ISR) address.
- The **INTA (Interrupt Acknowledge)** signal is sent to acknowledge the interrupt.

vii. Serial I/O and Control Signals

- The **Serial I/O Control** unit handles serial communication via the **SID (Serial Input Data)** and **SOD (Serial Output Data)** pins.

- Control signals like **READY**, **HOLD**, **HLDA (Hold Acknowledge)**, **S0, S1 (Status Signals)**, and **DMA** manage external device interactions, such as waiting for slow peripherals or handling direct memory access.

viii. Program Counter Update

- After the instruction is executed, the **Program Counter** is incremented (via the **Increment/Decrement Address Latch**) to point to the next instruction, and the cycle repeats.

Key Components and Their Roles

- **8-bit Internal Bus**: Facilitates data transfer between internal components.
- **Multiplexer**: Manages data flow between registers and the ALU.
- **Register Array**: Stores data and addresses for processing.
- **Timing & Control**: Generates control signals to orchestrate the fetch-decode-execute cycle.
- **Address/Data Bus**: Multiplexed bus for transferring addresses and data to/from external memory or I/O devices.

Summary of the Workflow

1. The 8085 starts with a clock signal and initializes via a reset.
2. It fetches an instruction from memory using the Program Counter and Address Bus.
3. The instruction is decoded, and operands are fetched if needed.
4. The ALU executes the instruction, updating flags and storing results.
5. Data is transferred to/from memory or I/O devices as required.
6. Interrupts are handled if triggered.
7. The Program Counter is updated, and the cycle repeats for the next instruction.

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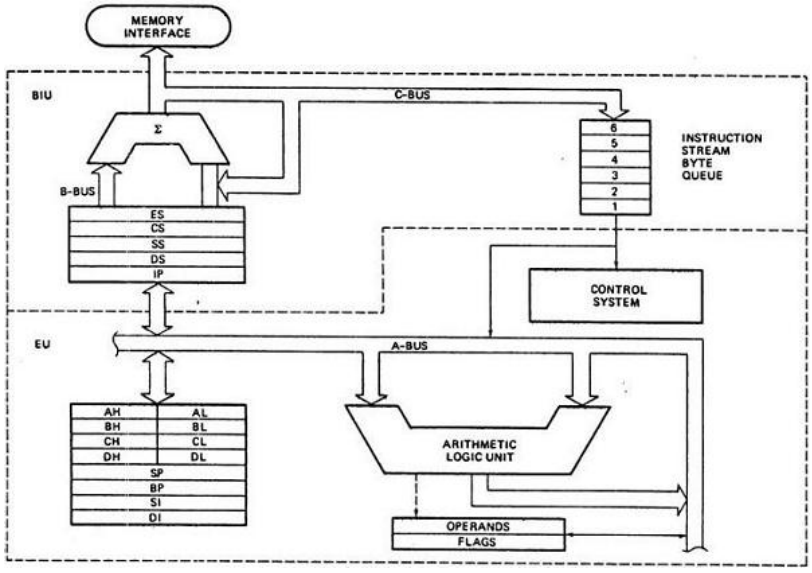
1. Overview of Microprocessors

A microprocessor is an integrated circuit that functions as the central processing unit (CPU) of a computer system, capable of fetching, decoding, and executing instructions. The Intel 8086, introduced in 1978, is a 16-bit microprocessor that marked a significant advancement over its 8-bit predecessors, such as the 8085. It became the foundation for the x86 architecture, widely used in modern computing systems, including personal computers.

2. Features of the 8086 Microprocessor

- **16-bit Data Bus**: The 8086 processes 16 bits of data at a time, doubling the data-handling capacity of the 8085.
- **20-bit Address Bus**: It can address up to 2^{20} = 1,048,576 memory locations (1 MB).
- **Clock Speed**: Typically operates at 5 MHz, with versions up to 10 MHz.
- **Power Supply**: Requires a single +5V DC power supply.
- **Instruction Set**: Supports a rich set of instructions (approximately 117 basic instructions), including multiplication and division.
- **Registers**: Includes four 16-bit general-purpose registers (AX, BX, CX, DX), four segment registers (CS, DS, SS, ES), a stack pointer (SP), base pointer (BP), source index (SI), destination index (DI), instruction pointer (IP), and a flags register.
- **Interrupts**: Supports 256 interrupt vectors, including hardware interrupts (NMI, INTR) and software interrupts.
- **Pipelining**: Introduces a basic form of instruction prefetching with a 6-byte instruction queue, improving execution speed.
- **Modes**: Operates in two modes—Minimum Mode (for small systems) and Maximum Mode (for multiprocessing systems).

3. Architecture of the 8086 Microprocessor



The 8086 architecture is divided into two main units: the **Bus Interface Unit (BIU)** and the **Execution Unit (EU)**, enabling parallel instruction fetching and execution.

3.1. Bus Interface Unit (BIU)

- Handles all interactions with memory and I/O devices.
- Contains:
  - **Segment Registers**: CS (Code Segment), DS (Data Segment), SS (Stack Segment), ES (Extra Segment) for memory segmentation.
  - **Instruction Pointer (IP)**: A 16-bit register that holds the offset address of the next instruction.
  - **Instruction Queue**: A 6-byte prefetch queue to store instructions ahead of execution.
  - **Address Generation Logic**: Combines segment addresses (shifted left by 4 bits) with offset addresses to form a 20-bit physical address.

3.2. Execution Unit (EU)

- Executes instructions fetched by the BIU.
- Contains:
  - **Arithmetic and Logic Unit (ALU)**: Performs 16-bit arithmetic (addition, subtraction, multiplication, division) and logical operations (AND, OR, XOR, NOT).
  - **Registers**:
    - **Accumulator (AX)**: Used for arithmetic, logic, and I/O operations (divided into AH and AL for 8-bit operations).
    - **Base (BX)**: Often used as a pointer to data (BH and BL for 8-bit).
    - **Counter (CX)**: Used in loops and shifts (CH and CL for 8-bit).
    - **Data (DX)**: Used in I/O and multiplication/division (DH and DL for 8-bit).
    - **Stack Pointer (SP)**: Points to the top of the stack.
    - **Base Pointer (BP)**: Used for stack-based addressing.
    - **Source Index (SI)** and **Destination Index (DI)**: Used for string operations and memory addressing.
  - **Flags Register**: Contains status flags (e.g., Zero, Sign, Carry, Overflow, Parity, Auxiliary Carry).

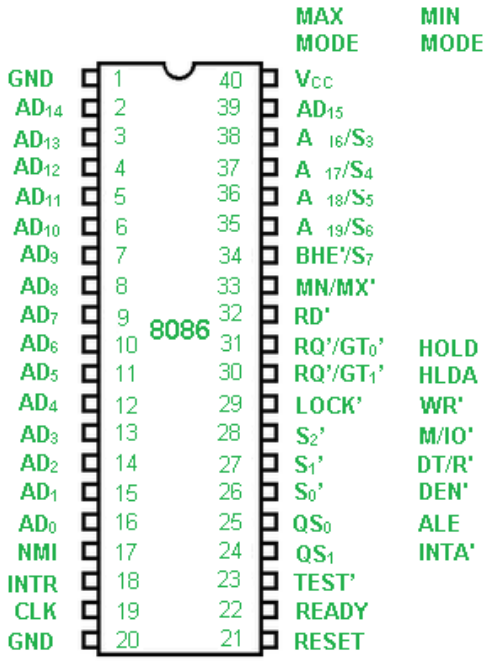
3.3. Control Unit

- Coordinates operations between the BIU and EU.
- Generates timing and control signals for instruction execution.

4. Pin Diagram of the 8086

The 8086 has 40 pins in a dual in-line package (DIP). Key pins include:

- **Address/Data Bus (AD0-AD15)**: Multiplexed 16-bit address and data lines.
- **Address/Status Bus (A16/S3-A19/S6)**: Higher-order address lines multiplexed with status signals.
- **Control Signals**:
  - **RD**: Read signal (active low).
  - **WR**: Write signal (active low).
  - **ALE**: Address Latch Enable to demultiplex address/data lines.
  - **M/IO**: Distinguishes between memory and I/O operations.
  - **BHE**: Bus High Enable for accessing the upper 8 bits of the data bus.
- **Interrupts**: NMI (Non-Maskable Interrupt), INTR (Interrupt Request), INTA (Interrupt Acknowledge).
- **Clock**: CLK pin for external clock input.
- **Mode Pins**: MN/MX to select Minimum or Maximum Mode.



5. Workflow of the 8086 Microprocessor

i. Power Supply and Clock Initialization

- The process begins with a +5V power supply and ground connection.
- An external clock signal is provided to the CLK pin, driving the internal timing and control unit.

ii. Instruction Fetch

- The BIU calculates the 20-bit physical address using a segment register (e.g., CS) and the Instruction Pointer (IP):
  Physical Address = Segment Address × 16 + Offset (IP).

- The address is placed on the Address/Data Bus (AD0-AD15) and Address/Status Bus (A16-A19), with ALE signaling a valid address.
- The BIU fetches up to 6 bytes of instructions into the instruction queue using the RD signal.

iii. Instruction Decode

- The EU retrieves the instruction from the queue and decodes it using the instruction decoder.
- Control signals are generated to determine the operation (e.g., arithmetic, data transfer, branching).

iv. Operand Fetch and Execution

- If operands are required, the BIU fetches them from memory or registers using segment:offset addressing.
- Operands are loaded into registers (e.g., AX, BX) or temporary internal storage.
- The ALU performs the operation, updating the Flags Register based on the result (e.g., Carry, Zero).

v. Data Transfer and Storage

- Results are stored in registers, memory, or I/O devices.
- For memory/I/O writes, the WR signal is asserted, and data is transferred via AD0-AD15.
- The Stack Pointer (SP) and Base Pointer (BP) manage stack operations (e.g., PUSH, POP).

vi. Interrupt Handling

- The 8086 supports 256 interrupt vectors (0–255).
- When an interrupt (e.g., INTR or NMI) occurs, the current IP and CS values are pushed onto the stack, and execution jumps to the interrupt service routine (ISR) address.
- INTA acknowledges the interrupt.

vii. Pipelining and Control Signals

- The BIU prefetches instructions while the EU executes the current instruction, improving efficiency.
- Control signals like READY, HOLD, and HLDA manage external device interactions (e.g., DMA).

viii. Instruction Pointer Update

- After execution, the Instruction Pointer (IP) is incremented to point to the next instruction, and the cycle repeats.

Key Components and Their Roles

- **16-bit Internal Bus**: Transfers data between registers, ALU, and other units.
- **Segment Registers**: Enable memory segmentation for accessing 1 MB of memory.
- **Instruction Queue**: Stores prefetched instructions for faster execution.
- **Timing & Control**: Synchronizes BIU and EU operations.
- **Address/Data Bus**: Multiplexed bus for address and data transfer.

Summary of the Workflow

1. The 8086 initializes with a clock signal and power supply.
2. The BIU fetches instructions into the queue using segment:offset addressing.
3. The EU decodes the instruction and fetches operands if needed.
4. The ALU executes the instruction, updating flags and storing results.
5. Data is transferred to/from memory or I/O as required.
6. Interrupts are processed if triggered.
7. The Instruction Pointer is updated, and the cycle repeats.

Comparison Between 8085 and 8086 Microprocessors

Feature	8085 Microprocessor	8086 Microprocessor
Introduced By	Intel (1976)	Intel (1978)
Architecture	8-bit	16-bit
Data Bus Width	8-bit	16-bit
Address Bus Width	16-bit (can address 64 KB memory)	20-bit (can address 1 MB memory)
Instruction Set	Simple, fewer instructions	More powerful, includes string operations and multiplication/division
Number of Registers	Limited general-purpose registers	More general-purpose and segment registers
Memory Segmentation	No segmentation	Supports segmentation (CS, DS, SS, ES)
Operating Mode	Only real mode	Real mode and protected mode
Execution Speed	Slower due to 8-bit operations	Faster with 16-bit processing
Clock Speed	Typically 3 MHz	Typically 5 MHz to 10 MHz
Addressing Modes	Limited addressing modes	More addressing modes (Register, Immediate, Direct, Indirect, Indexed, etc.)
ALU Size	8-bit	16-bit
Multiplication & Division	No direct support	Hardware-supported multiplication & division
Pipeline	No instruction queue (fetch-execute cycle)	6-byte instruction queue (fetch and execute simultaneously)
I/O Operations	8-bit I/O operations	16-bit I/O operations
Co-Processor Support	No support	Supports 8087 Math Co-Processor

Number of Pins	40 pins	40 pins (minimum mode) / 68 pins (maximum mode)
Mode of Operation	Single-mode	Minimum mode (like 8085) & Maximum mode (for multiprocessor system)
Interrupts	5 hardware interrupts	256 interrupt types